

EMERGING TECHNOLOGIES: OVERVIEW, APPLICATIONS, AND SOCIETAL IMPACTS

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Technologies

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I. Introduction

Emerging technologies are technologies that are still developing, improving, or being applied in new ways and that have the potential to change how people, organizations, and industries operate. Technology does not have to be completely new to be considered emerging. An existing technology may also become emerging when it is combined with other technologies or applied to a new purpose. Artificial intelligence is a good example because it began decades ago but continues to develop and appear in new research areas and applications.

For a technology analyst, understanding emergence is important because technologies can create opportunities while also producing uncertainty. Technology may grow quickly and have a large future impact, but its final applications and consequences may not yet be fully known. Therefore, emerging technology should be evaluated not only according to what it can do today but also according to how it may develop and affect society in the future.

II. Characteristics of Emerging Technology

According to Rotolo, Hicks, and Martin (2015), there are five characteristics that help explain when a technology can be considered emerging.

Characteristic	Explanation	Example
Radical Novelty	Technology may be genuinely new, or existing technology may be used according to a new principle or for a new purpose.	Artificial intelligence can be adapted for different applications to achieve different results.
Relatively fast growth	Emerging technologies tend to expand quickly in areas such as users, research, funding, prototypes, products, and services.	AI is increasingly used by companies, universities, and individuals.

Coherence	An emerging technology can result from the convergence of previously separate technologies or research areas that have developed beyond the purely conceptual stage.	Cloud computing, AI, data analytics, and robotics can work together for different uses.
Prominent impact	The technology has the potential to produce noticeable changes in industries, institutions, and society.	AI can affect business, education, healthcare, and individuals.
Uncertainty and ambiguity	The future effects, applications, and outcomes are not completely predictable. Some proposed uses may not succeed, and different groups may interpret the technology differently.	Some possible AI applications can be predicted, but their successful development and eventual effects remain uncertain.

III. Areas of Emerging Technology

There are several important areas of emerging technology. These technologies differ in their functions, but they can also complement one another and become part of larger technological systems.

1. Augmented Reality (AR)

Augmented Reality enhances a person's view of the physical world by placing computer-generated information, sounds, text, or visual effects into that real-world view. Instead of completely replacing the surroundings, AR adds digital information to them. Several examples include: an in-car display can guide a driver and assist with safe reverse parking; Pokémon GO uses AR in gaming;

and furniture applications can allow shoppers to preview how an item might look in a room before purchasing it.

2. Artificial Intelligence (AI)

Artificial Intelligence refers to the simulation of human intelligence by machines. It is considered emerging because its capabilities and applications continue to expand. Its influence can reach businesses, academic institutions, healthcare, and individuals. It has grown in use, can be combined with other technologies, can produce significant changes, and still has uncertain future applications.

3. Machine Learning

Machine learning is an application of AI in which a machine learns from experience or data in a way that resembles aspects of human learning. It can be applied in healthcare, financial services, marketing, and transportation. In healthcare, for example, algorithms can assist efforts to identify and develop new medicines more rapidly. This illustrates how technology can move from a general concept into specialized applications in different sectors.

4. Quantum Computers

Quantum computers use principles of quantum mechanics to perform certain calculations more efficiently than conventional computers. Their basic unit is the qubit. Unlike an ordinary bit that represents either 0 or 1, a qubit can exist in a quantum superposition of computational states.

IBM, Google, Rigetti Computing, and IonQ are organizations demonstrating quantum computers. Researchers and academics are also developing quantum programs and software. If the technology becomes sufficiently refined, powerful quantum computers could be used to simulate aspects of nature, including atoms and molecules, and could support the design of new materials.

IV. Industrial Revolution

The Industrial Revolution describes major stages in the transformation of production technology. Each stage introduced technologies that changed manufacturing, working conditions, and people's lifestyles.

Stage	Period	Key Characteristics
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Industry 1.0	18th century	Mechanical production using water/hydroelectric and steam-powered equipment.
Industry 2.0	19th century	Electricity enabled greater production and more sophisticated machines.
Industry 3.0	Computer era	Electronics, computers, information technology, connectivity, and increased automation transformed production.
Industry 4.0	Current and developing environment	Connected and intelligent technologies such as IoT, robotics, VR, AR, and AI change production and relationships among people, machines, suppliers, producers, and customers.

V. Building Blocks of Industry 4.0

Industry 4.0 is supported by several technologies that connect physical systems, data, software, and people.

1. Big Data Analytics

Big Data Analytics involves collecting, organizing, and examining very large amounts of data from different sources to identify patterns and useful information. In production and customer management, it can support real-time decisions. The major value of this technology is that large datasets can be transformed into information that helps organizations respond more quickly.

2. Internet of Things (IoT)

The Internet of Things refers to physical objects connected to the Internet so they can communicate and interact with other devices. Sensors and wireless technologies can allow field devices to communicate with centralized controllers. IoT can distribute analytics and decision-making closer to the devices, allowing systems to produce responses and results in real time.

3. Cloud Computing

Cloud computing is the delivery of hosted services through the Internet. The lecture explains that improving cloud performance can support very fast response times. Machine data and functionality can be placed in the cloud, enabling more data-driven services in production systems.

4. Augmented Reality (AR) in Industry 4.0

AR can support workers by presenting useful information while they perform tasks. As it becomes more sophisticated, organizations may use it more broadly to provide workers with real-time information, helping improve decisions and work procedures.

VI. Analysis

Technology should not be viewed as separate. Their greatest value can occur when they are combined. For example, IoT devices can collect information, Big Data Analytics can organize and analyze that information, cloud computing can provide an environment for hosting data and services, and AI or machine learning can support intelligent analysis. AR can then present useful information to a worker through a real-world view.

This interconnected approach reflects the coherence characteristic of emerging technology and the direction of Industry 4.0. It also demonstrates why emerging technologies can have prominent impacts across multiple sectors rather than remaining limited to one field.

VII. Effects of Emerging Technology on Society

Emerging technology can create substantial benefits for society but also emphasizes uncertainty and ambiguity. Its effects can therefore be understood as both opportunities and challenges.

Positive effects include improved efficiency, faster access to information, better decision-making, new services, and advances in fields such as healthcare. AI and machine learning can support healthcare research, while cloud computing and connected systems can improve access to data and services. AR can provide useful information in work, transportation, shopping, and entertainment. Industry 4.0 can also change production by increasing automation and connectivity.

At the same time, emerging technology can create uncertain outcomes. Some proposed applications may not develop successfully, while others may produce unintended effects. Because technologies can change relationships among people, organizations, and machines, society must consider how these changes will affect future practices. Therefore, rather than assuming that every new technology will have only positive outcomes, uncertainty and ambiguity are also significant aspects of technological emergence.

VIII. Augmented Reality (AR) vs Virtual Reality (VR)

Augmented Reality and Virtual Reality both involve computer-generated experiences, but they interact with the user's environment in different ways. AR adds digital content to the real world. The user can still see the physical surroundings while receiving additional digital information.

Virtual Reality, in contrast, creates a computer-generated environment that the user experiences as a virtual space. Rather than simply adding digital information to the real surroundings, VR is designed to immerse the user in a simulated environment. Therefore, the simplest distinction is: AR enhances the real world, while VR provides a virtual environment.

IX. Advantages and Disadvantages of Artificial Intelligence

AI can provide important benefits, but its emerging nature also means that its effects should be considered carefully.

Advantage	Explanation
Faster processing	AI can perform certain information-processing tasks quickly, which can support faster work and analysis.
Automation	AI can assist with tasks that can be performed by machines, helping improve efficiency.

Data-based support	AI can help process information and support decisions, particularly when large amounts of data are involved.
Healthcare applications	The lecture identifies healthcare as an area where AI and machine learning can contribute to research and medicine development.
Wide range of applications	AI can be applied in businesses, academe, healthcare, and other areas, giving it broad potential impact.

Disadvantage	Explanation
Uncertain outcomes	The lecture stresses that future AI applications and their results cannot be predicted with complete certainty.
Unintended effects	Emerging technologies may produce outcomes that were not originally intended or expected.
Ambiguity of use	Different groups may understand or use the same technology in different ways, creating uncertainty about its role.
Dependence on developing systems	Because AI continues to develop, some proposed applications may remain experimental or may not become successful.
Broad societal impact	AI can affect many sectors and institutions, so changes in how people and organizations interact may be significant and difficult to predict.

X. Recommendations for Technology Adoption

Organizations considering emerging technology should evaluate both current capability and future uncertainty. First, they should identify the problem or process that the technology is intended to improve. Second, they should examine whether technology has demonstrated growth and coherence with other systems. Third, they should consider its potential impact on workers, customers, institutions, and production processes. Finally, organizations should recognize that emerging technologies may change over time and that some expected applications may not develop as planned.

For an organization moving toward Industry 4.0, a connected approach can be considered. IoT can provide data from physical devices, Big Data Analytics can

examine the collected information, cloud computing can support hosted services and data-driven operations, and AI can assist with learning and analysis. AR can provide workers with real-time information when appropriate. The goal is not simply to adopt the newest technology, but to understand how technologies can work together to solve real problems.

XI. Conclusion

Emerging technologies are characterized by novelty, rapid growth, coherence, potential impact, and uncertainty. Technologies such as AR, AI, machine learning, and quantum computing can continue to be considered emerging because their capabilities and applications are still developing. The Fourth Industrial Revolution demonstrates how these technologies can become part of connected production environments through tools such as Big Data Analytics, IoT, cloud computing, and AR.

The major lesson is that emerging technology is not defined only by being new. Its importance also comes from how it develops, connects with other technologies, affects society, and creates possibilities that may not yet be fully understood. For future IT professionals, understanding these characteristics provides a foundation for evaluating technology responsibly and recognizing both its opportunities and uncertainties.

XII. References

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- Rotolo, D., Hicks, D., & Martin, B. (2015). What is an emerging technology? SPRU Working Paper Series, 44, 1–34.
- TechTarget. (n.d.). Artificial intelligence.
- TechTarget. (n.d.). Cloud computing.
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XIII. Required Checking Results

1. Proof of Grammar Checking Using Grammarly

The screenshot shows the Grammarly interface with a document titled "Module 1 Activity 1 Barrientos.docx". A "Performance" pop-up window is displayed, showing a text score of 91 out of 100. The window also displays word count (10,629 characters, 1,491 words, 118 sentences) and readability metrics (word length 5.9, sentence length 12.6, readability score 25). The background shows the document text, which includes an introduction to emerging technologies.

Performance

Text score: 91 out of 100. This score represents the quality of writing in this document. You can increase it by addressing Grammarly's suggestions.

Word count

Metric	Value	Reading time	Speaking time
Characters	10,629	5 min 57 sec	
Words	1,491		11 min 28 sec
Sentences	118		

Readability

Metrics compared to other Grammarly users

Metric	Value	Comparison
Word length	5.9	Above average
Sentence length	12.6	Above average
Readability score	25	

Your text is likely to be understood by college graduates but may not be easy for many to read.

2. Proof Of AI Checking Using Pangram

The screenshot shows the Pangram AI Detection Dashboard. The document "Module 1 - Activity 1 - Barrientos.docx" is analyzed, showing a 63% AI detection rate. The dashboard includes a sidebar with navigation options (Text Detection, Image Detection, History, All Checks, More Features) and a main content area with an "Overview" tab. The "Overview" tab displays a circular progress chart showing the AI detection results.

Module 1 - Activity 1 - Barrientos.docx

Sep 11, 2026 12:00 AM | 1,953 words

AI Highlight

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AI Detected

1,953 words scanned

We believe that this text is a mix of AI, AI-assisted, and human-written content.

63% of this text is AI

AI 63% | AI Assisted 31% | Human 6%